

TOAN NGUYEN DOAN

AI Engineer Intern

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Education

HUS - Hanoi University of Science, VNU
B.S. Data Science.

Hanoi, Vietnam
Expected Graduation: 2027

Skills

Programming Languages: Python, R, SQL, Java, C++.

AI & Machine Learning: Computer Vision, Data Preprocessing & Cleaning, Statistical Modeling.

Frameworks & Libraries: Scikit-learn, Pandas, NumPy, Matplotlib.

Tools & Cloud: Jupyter Notebook, Docker, GitHub, Ngrok.

Concepts: Data Structures & Algorithms, Model Optimization & Evaluation, RESTful APIs, OOP, Linear Programming, Numerical Analysis.

Experience

Undergraduate Research Assistant

Jan 2026 – Jun 2026

Supervised by Dr. Thuy - Hanoi University of Science, VNU

- Assisted in developing statistical models and time-series forecasting pipelines utilizing **ARMA/GARCH** to analyze data trends.
- Programmed optimization solvers to resolve complex linear programming problems utilizing **Simplex** and **Big-M** methods.
- Applied **Bayesian inference** and **Monte-Carlo** simulations utilizing **R** to evaluate probabilistic models and algorithm performance.

Projects

Ship Room Monitoring – Scientific Research Project

An intelligent computer vision system to track people and packages in delivery rooms.

- Implemented object detection and multi-object tracking pipelines using **YOLOv8** and **DeepSORT** algorithm to consistently monitor bounding boxes across video frames.
- Configured custom polygon tracking zones (e.g., `[[1060, 520], [1480, 520], [1480, 860], [1060, 860]]`) utilizing the **Supervision** library to detect interactions and movements within specifically defined coordinate areas.
- Deployed the AI models on a cloud environment via **Google Colab** and configured **Ngrok** authentication to enable remote real-time video streaming.

AI Traffic Camera Monitoring – Personal Project

A computer vision application for real-time vehicle detection and traffic flow analysis.

- Developed an object detection model using **YOLOv8** to identify and classify various vehicle types in diverse traffic conditions.
- Applied **OpenCV** for image processing and bounding box rendering to extract traffic insights.
- GitHub:** github.com/dekii2275/CV_trafficdetection

Airline Passenger Satisfaction Classification – Machine Learning Project

An end-to-end machine learning pipeline to predict customer satisfaction levels.

- Performed extensive **Exploratory Data Analysis (EDA)** and data preprocessing, including handling missing values, encoding categorical features, and feature scaling using **Pandas** and **Scikit-learn**.
- Trained and evaluated multiple classification models (**KNN**, **Random Forest**, **SVM**) to identify the most critical factors influencing passenger satisfaction.

ZEnergy - Energy Management System – Frontend Project

A web interface dedicated to displaying and managing energy usage information.

- Developed an interactive and responsive frontend interface using **HTML**, **CSS**, **JavaScript** to visualize energy consumption data.
- Designed intuitive data dashboards to help users efficiently monitor and manage their energy usage metrics.
- GitHub:** github.com/dekii2275/oilandenegry_website

Certifications

SIC AI - Samsung Innovation Campus 2025

2025